

1 Let a be a positive constant.

- (a) Use the method of differences to find $\sum_{r=1}^n \frac{1}{(ar+1)(ar+a+1)}$ in terms of n and a . [4]

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- (b) Find the value of a for which $\sum_{r=1}^{\infty} \frac{1}{(ar+1)(ar+a+1)} = \frac{1}{6}$. [3]

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